AD 502 – Machine Learning

Pre-Reading Material – Lecture 35

Basic Ensemble Techniques: Weighted Average

Estimated Reading Time: 5 minutes

Introduction

Ensemble Learning is a machine learning technique in which multiple models are combined to produce more reliable predictions. Weighted Average is an ensemble technique in which different weights are assigned to individual models according to their predictive performance, reliability, or importance.

Unlike simple averaging, where all models contribute equally, weighted averaging allows selected models to have a greater influence on the final prediction.

Working of Weighted Average

The weighted average technique involves training multiple machine learning models, generating predictions from each model, assigning appropriate weights, and combining their predictions to obtain the final output.

Models that perform more reliably may be assigned higher weights, while models with lower predictive performance may receive lower weights. The weights can be selected using validation data or domain knowledge.

Advantages

- Gives greater influence to more reliable models.
- Combines the strengths of different machine learning models.
- Can improve prediction accuracy when models provide complementary information.
- Provides flexibility in combining model predictions.

Limitations

- Selecting appropriate weights can be challenging.

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- Poor weight selection may reduce prediction accuracy.
- Performance improvement is not guaranteed.
- Evaluating multiple models requires additional computational resources.

Applications

Weighted average is used in weather forecasting, electricity demand prediction, renewable energy forecasting, financial forecasting, sales prediction, and other applications where combining predictions from multiple models can improve reliability.

Key Takeaway

Weighted Average is an ensemble learning technique that combines predictions from multiple models by assigning different levels of importance to their outputs. Appropriate weight selection and validation are essential for obtaining reliable predictions.

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